

An Alternative to SubGHz for LPWAN

Swen Leugner

Center of Excellence CoSA
University of Applied Science Lübeck
Lübeck, Germany
swen.leugner@th-luebeck.de

Horst Hellbrück

Center of Excellence CoSA
University of Applied Science Lübeck
Lübeck, Germany
horst.hellbrueck@th-luebeck.de

Abstract—Low Power Wide Area Networks (LPWAN) enable energy-efficient communication over long distances, making them interesting for Internet of Things (IoT) applications. Some applications require a higher data rate than standard LPWAN while keeping a moderate range for keeping infrastructure requirements low. This paper provides a comparative evaluation of two LPWAN technologies, LoRa operating at 2.4 GHz and DECT NR+ for these kind of applications. We conducted modeling and experimental evaluations to assess their performance in terms of range and data rate across indoor, indoor-outdoor, and outdoor scenarios. For modeling we applied the log-distance path loss model, accounting for environmental conditions. Results show that LoRa achieves significantly greater range, with up to 259 meters in indoor-outdoor scenarios, compared to DECT NR+ with a maximum of 16 meters. Outdoor evaluations revealed LoRa's capability to maintain connectivity at distances exceeding 1 kilometer in non-line-of-sight conditions. However, DECT NR+ offers higher data rates, reaching up to 3744 kBit/s, making it advantageous for applications demanding robust data rates in limited-range scenarios. The findings highlight the trade-offs between range and data rate.

Index Terms—IoT, LPWAN, LoRa, DECT NR+, ISM, RF Modeling

I. INTRODUCTION

The Internet of Things (IoT) is rapidly evolving, transforming the way we live and work. Modern devices are increasingly equipped with wireless radio interfaces, enabling seamless communication. In the context of smart city projects and industrial applications, the number of wireless sensors operating within the 868 MHz band has significantly grown in the last years. This trend demonstrates the importance of Low Power Wide Area Networks (LPWANs) for IoT solutions, with LoRaWAN being a prime solution for energy efficiency and extended range.

LoRaWAN in the 868 MHz band offers key advantages, including excellent building penetration, long battery life, and extended range [1] [2]. These characteristics are particularly crucial for IoT applications like water meter readings [3]. However, its low data rate and regulatory restrictions, such as the 1% duty cycle limitation in Europe, pose challenges to dynamic control of connected actuators [4]. Such constraints limit its potential for real-time applications that require bidirectional communication, such as those vital for emergency services like firefighters.

Emerging technologies, such as DECT NR+ and LoRa in the 2.4 GHz band, provide viable alternatives to the limitations

of the 868 MHz band. LoRa operating in the global 2.4 GHz ISM band is free from duty cycle restrictions, while DECT NR+ operates within the 1.88–1.9 GHz band in Europe. This exclusive frequency allocation prevents interference with other radio standards, thereby reducing transmission collisions and ensuring higher reliability, particularly in urban or industrial settings.

Selecting the appropriate radio standard strongly depends on the specific application requirements. Parameters such as range, data rate, and building penetration must be balanced to suit diverse scenarios, ranging from building control and industrial automation to critical emergency communication. However, a systematic analysis comparing DECT NR+ and LoRa in the 2.4 GHz band remain scarce.

This publication aims to address this gap by providing a comprehensive evaluation of the two technologies. Combining modeling and experimental validation, this study examines range, building penetration, and data rate. The results aim to offer technical insights and practical recommendations for tailored IoT applications.

The contributions of our work are as follows:

- Comparison between DECT NR+ and LoRa 2.4 GHz
- Modeling and evaluation of realistic scenarios
- Recommendations for practical applications
- New contribution to the discussion on LPWAN technologies

The rest of the paper is organized as follows. Section II discusses related work for DECT NR+ and LoRa 2.4 GHz. In Section III, we introduce the modeling framework and methodology used in this study. We provide an evaluation of range and performance in Section IV, presenting the experimental findings. Finally, we draw conclusions and suggest directions for future work in Section V.

II. RELATED WORK

In this section, we will present related work on LoRa in the 2.4 GHz frequency band and DECT NR+. We will start with DECT NR+ first. Various terms are often used for DECT NR+ in the literature. It is sometimes called DECT-2020 NR, DECT NR or similar. We will continue to use the term DECT NR+ for this paper, but it is mentioned for the sake of completeness. In [5], Samuylov et al. investigate coexistence mechanisms and the performance of DECT NR+ compared to classic DECT. They use a simulation at system level and

come to the conclusion that DECT NR+ is coexistent with classic DECT. In [6] Maliatsos et al. briefly introduce DECT NR+, they show that DECT NR+ uses a cluster tree topology to build a mesh network and emphasize this feature in DECT NR+. In [7], Troccoli et al. present a new type of localization in DECT NR+. The authors use direction-of-arrival to enable localization in large DECT NR+ networks. The direction-of-arrival is implemented in Matlab as a simulation environment. The solution is not implemented or evaluated in or with DECT NR+ hardware. In [8], Waßmann et al. carry out a performance evaluation with DECT NR+ in an industrial environment. The authors created DECT NR+ frames with Matlab and used software-defined radios for transmission and reception. The authors examined the 450 MHz, 1350 MHz and 3700 MHz frequency bands. The 450 MHz and 3700 MHz frequency bands are subject to licensing in Germany. The 1350 MHz band is only released for wireless microphones in Germany. The 1880-1900 MHz frequency band intended exclusively for classic DECT or DECT NR+ was not evaluated.

In [9] Haque et al. evaluated the link distance performance in different radio environments. To perform the evaluation, they used the free space pathloss model, 3GPP TR 38.901 path loss model, two-ray ground reflection model and COST-231 hata model. The models all showed mixed results. We will therefore use the log-distance path loss model for our investigation, as it can be easily adapted to other environments using the path loss exponent. In addition to modeling the path loss models, Haque et al. also carried out measurements with real hardware. The authors were able to achieve ranges of up to 190 m indoors. Outdoor ranges of up to 2.4 km could be achieved. However, the range was not estimated using a model. The data rate was also not specified for the maximum distances that can be achieved between the transmitter and receiver. However, this is an important criterion for assessing whether the radio standard is suitable for the user's intended application. We will therefore add the data rate to the model in our publication.

In [10], Janssen et al. discussed LoRa 2.4 GHz and specified the range and data rate using the free space and dominated path loss model. They described unusually long distances (up to 133 km). In addition to the model, Janssen et al. did not carry out any validation in a real radio environment in which the unusually long radio environment would have been noticeable. In [11], Deréviackine et al. compared LoRa in the 2.4 GHz frequency band with LoRa in the 868 MHz band. They carried out indoor and outdoor experiments for both frequency bands with the SX1276 radio chip and SX1280 radio chip from Semtech, but did not develop a model. Deréviackine et al. see several advantages in the 2.4 GHz frequency band. On the one hand, LoRa in the 2.4 GHz frequency band achieves higher data rates than LoRa in the 868 MHz frequency band. Secondly, the radio channel is better utilized as there is no duty cycle limitation in the 2.4 GHz frequency band. For example, in Europe only 1% of the 868 MHz may be occupied by the respective 868 MHz LoRa device. With regard to robustness against interference, Deréviackine et al. state that LoRa

modulation can also cause demodulation, although interference from other technologies is present.

In our research, we were unable to find a direct comparison between DECT NR+ and LoRa 2.4 GHz. Furthermore, the related work shows that unusually long or short distances are assumed for DECT NR+ and LoRa 2.4 GHz. We will therefore first carry out a modeling of the two radio standards in order to compare them. We will then carry out a range evaluation.

III. MODELING

In order to show the differences between LoRa and DECT NR+, we will first carry out a modeling. The aim of the modeling is to obtain an estimate of the range and data rate of the two radio standards in order to be able to identify applications for them. In the next section, we will then empirically determine the range in a test.

For modeling purposes, we use a transmitter, a transmission channel and a receiver. The transmitter emits a continuous wave, which is attenuated via the transmission channel and received by the receiver. To model the attenuation, we use the log-distance path loss model. The log-distance path loss model is suitable for the frequency range from 1.9 GHz to 2.5 GHz and can be easily adapted for urban, suburban or rural areas. The log-distance path loss model is calculated as

$$PL(d) = PL(d_0) + 10 \cdot n \cdot \log_{10} \left(\frac{d}{d_0} \right) + 20 \cdot \log_{10} \left(\frac{f}{1000} \right) + X_\sigma, \quad (1)$$

where d represents the distance between the transmitter and the receiver in meters, while d_0 is the reference distance at which the path loss, $PL(d_0)$, is known. The term $PL(d_0)$ (in dB) stands for the path loss at the reference distance d_0 . The path loss exponent, n , characterizes the attenuation of the signal and depends on the environmental conditions, such as whether the signal propagates in urban, rural, or indoor settings. The parameter X_σ accounts for shadowing effects modeled as a Gaussian random variable. Finally, f refers to the carrier frequency of the signal, expressed in megahertz.

The received signal power at the receiver at distance d is then calculated from

$$P(d)_{Rx} = P_{Tx} + G_{Tx} - PL(d) + G_{Rx}, \quad (2)$$

where P_{Tx} is the transmission power in dBm, G_{Tx} is the antenna gain in dB of the transmitter, $PL(d)$ is the path loss at distance d and G_{Rx} is the antenna gain in dB of the receiver.

To calculate the data rate for the received signal strength, we applied Table I. The values for LoRa were taken from the data sheet of the SX1280 LoRa transceiver [12]. The values for DECT NR+ were taken from Signal Vortex [13].

The table is structured in such a way that we have a column for LoRa and DECT NR+ radio technology. For LoRa, we have specified the data rate for the bandwidths 203 kHz and 1625 kHz and thus the smallest bandwidth and largest bandwidth of the SX1280 radio chip. For 31.72 kBit/s, the received signal must therefore have a received signal strength (RSS) of at least -109 dBm. This is analogous to the reception sensitivity that is achieved with a spreading factor of 5.

TABLE I
DATA RATES FOR DIFFERENT MODES BASED ON RSS VALUES.

Technology	RSS Range (dBm)	Data Rate (kBit/s)
LoRa 203 kHz	≥ -109	31.72
	$-111 \leq \text{RSS} < -109$	19.03
	$-115 \leq \text{RSS} < -111$	11.1
	$-118 \leq \text{RSS} < -115$	6.34
	$-121 \leq \text{RSS} < -118$	3.57
	$-124 \leq \text{RSS} < -121$	1.98
	$-130 \leq \text{RSS} < -124$	1.09
LoRa 1625 kHz	< -130	0
	≥ -99	253.91
	$-103 \leq \text{RSS} < -99$	152.34
	$-106 \leq \text{RSS} < -103$	88.87
	$-109 \leq \text{RSS} < -106$	50.78
	$-111 \leq \text{RSS} < -109$	28.56
	$-114 \leq \text{RSS} < -111$	15.87
DECT NR+	$-120 \leq \text{RSS} < -114$	8.73
	< -120	0
	≥ -81	3744
	$-85 \leq \text{RSS} < -81$	2496
	$-88 \leq \text{RSS} < -85$	1872
	$-90 \leq \text{RSS} < -88$	1248
	$-93 \leq \text{RSS} < -90$	624
	< -93	0

Accordingly, signals with a spreading factor of 12 can be received up to -130 dBm.

The minimum reception sensitivity for the modulation coding scheme were also assumed for DECT NR. MCS 4 is therefore used above -81 dBm. Signals can be received with MCS 0 to -93 dBm.

In the following, we will model a scenario in which the transmitter is indoors and the receiver is outdoors. We refer to this scenario as indoor-outdoor. As a further scenario, the transmitter is placed indoors and the receiver is also indoors. We refer to this scenario as indoor-indoor.

Table II shows the parameters used for the modeled scenarios.

TABLE II
PARAMETERS FOR INDOOR-INDOOR AND INDOOR-OUTDOOR SCENARIOS.

Parameter	Indoor-Indoor	Indoor-Outdoor
d_0	1.0 m	1.0 m
$PL(d_0)$	40.0 dB	40.0 dB
n	4.45	5.8
X_σ	0	0
f_{LoRa}	2400 MHz	2400 MHz
f_{DECT}	1900 MHz	1900 MHz
$P_{\text{TX, LoRa}}$	15 dBm	15 dBm
$P_{\text{TX, DECT, EU}}$	19 dBm	19 dBm
LoRa $G_{\text{TX}}, G_{\text{TX}}$	5 dBi	5 dBi
DECT NR+ $G_{\text{TX}}, G_{\text{TX}}$	4 dBi	4 dBi

We determined the path loss at $PL(d_0)$ empirically for the development kits used later at a distance of 1 m. The X_σ was assumed to be 0, as no Gaussian noise should be superimposed. This is a simplification to make the curves easier to read. The remaining parameters correspond to the development kits used later.

Figure 1 shows the indoor-indoor scenario. We see LoRa with 203 kHz bandwidth shown in blue, LoRa with 1625 kHz

bandwidth shown in orange and DECT NR+ shown in green. We can see that DECT NR+ has the shortest range but the highest data rate. The range in the model for DECT NR+ is 16 m. The range for LoRa with 203 kHz bandwidth is 70 m. The steps indicate the spreading factor (SF) for LoRa and the MCS for DECT NR+. The highest range with DECT NR+ is achieved with MCS0 and with LoRa with SF12.

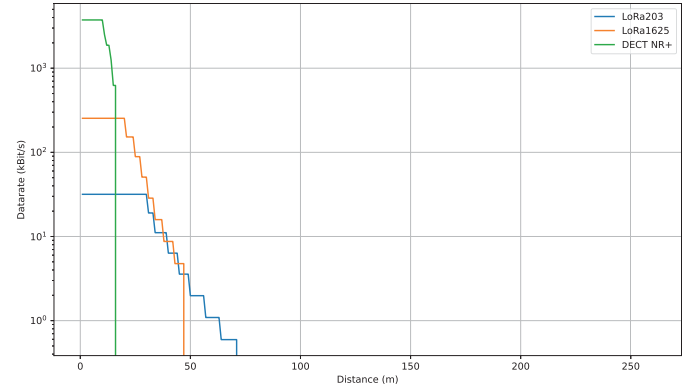


Fig. 1. Model Datarate vs Distance of LoRa and DECT NR+ for Indoor-Indoor Scenario

Figure 2 shows the indoor-outdoor scenario. The coloring of the curves is the same as in Figure 1. The same applies to the scaling of the X-axis. We therefore assume that a higher range is achieved in the indoor-outdoor scenario. In the model we assume 270 m. In principle, both scenarios show that DECT NR+ will have a lower range. It is about 2/3 below the range we expect with LoRa.

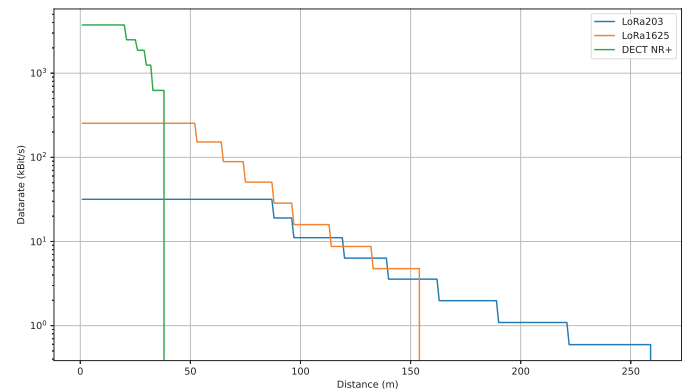


Fig. 2. Model Datarate vs Distance of LoRa and DECT NR+ for Indoor-Outdoor Scenario

However, if the range with DECT NR+ is sufficient, the data rate is higher than that of LoRa in all cases. We therefore deduce from the model: If the range is sufficient for the application, DECT NR+ should be preferred due to the higher data rate. In the next section, we will evaluate LoRa and DECT NR+ using development kits. The range evaluation will be carried out for this purpose. Due to the limited range of DECT NR+, we will primarily carry out the range evaluation with LoRa.

IV. RANGE EVALUATION

We are using two development kits to evaluate the range. The LoRa development kit is the T3S3 from Lillygo. It was chosen because, unlike the development kit from Semtech, it has a power amplifier. The power amplifier enables a total transmission power of +20 dBm, which is the maximum transmission power permitted in Germany in the 2.4 GHz band. The transceiver in the development kit is an SX1280 from Semtech. We used the DECT NR+ Evaluation Board from RFmondial as the DECT NR+ Development Kit. It has the NRF9161 transceiver from Nordicsemiconductor. A Raspberry PI Computation Module is installed on the evaluation board, which serves as an application processor. In both cases, only the physical layer is considered. A datalink layer is not considered. Figure 3 shows the development kits. The development kits are shown with the antennas included in the development kits.

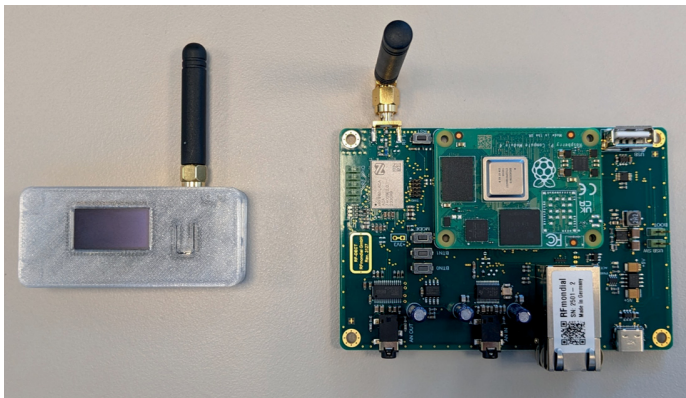


Fig. 3. Development kits Lillygo T3S3 (left), RFmondial DECT NR+ Evaluation Board (right)

The settings for the range evaluation of the individual kits are shown in Table III.

TABLE III
SETTINGS FOR RANGE EVALUATION

	LoRa	DECT NR+
Center Frequency	2400.5 MHz	1881.792 MHz
Bandwidth	203 kHz	1.728 MHz
TX Power	20 dBm	19 dBm
Modulation	LoRa SF12	MCS0

For LoRa we have chosen the lower 2400.5 GHz band as it is likely to interfere with WiFi. A bandwidth of 203 kHz was chosen to allow a long range. The transmission power (incl. antenna gain) was set to +20 dBm in order to transmit at the maximum permitted transmission power in Germany. As modulation we utilized LoRa with spreading factor (SF) 12. For DECT NR+, we have chosen the frequency 1881.792 MHz, which can be selected at will as DECT NR+ coexists with classic DECT. The bandwidth is fixed for the NRF9161 transceiver. The transmission power of +19 dBm is the maximum supported by the NRF9161 transceiver. The Modulation Scheme Index (MCS) 0 is the smallest that could

be set in the development kit. Both antennas have the same antenna gain of 4 dBi. For the evaluation, we configured one of the development kits as a transmitter and the other as a receiver.

For the further procedure, we first carried out an indoor-indoor test, which serves as a reference in order to carry out a subsequent outdoor test and thus determine the maximum range. The indoor-indoor test was carried out in Building 17, Building 18 and Building 13 of the University of Applied Science Lübeck. The buildings are marked in Figure 4. The buildings are multi-storey and have large glass panes. Inside there are laboratories, offices and seminar rooms, the partition walls are made of plasterboard. The transmitter was placed in the center of building 18 on the 2nd floor. The receiver was used to determine the maximum distance at which reception is still possible. This showed that DECT NR+ no longer allows reception after 16 meters. On the second floor, data could still be received within a range of 10 meters around the transmitter. On the first floor, reception was possible 4 meters around the transmitter.



Fig. 4. Evaluation area. Red marking for Indoor area. Red star for maximum indoor-outdoor range. Yellow marking as outdoor route

With the LoRa Development Kit, Building 17 and Building 18 could be covered across all floors. The signal was interrupted on the first floor of building 13. Data could be received within a radius of 59 m in that indoor-outdoor-indoor test. The maximum range between indoor transmitter and outdoor receiver was 259 m and is marked a red star in Figure 4. As the measurements corresponded to the indoor-outdoor model in relation to each other, the outdoor-outdoor investigations were carried out exclusively with LoRa. The measurement path is marked in yellow in Figure 4.

For the outdoor evaluation, the transmitter was placed outside the building at a height of 10 m in the corridor between Building 17 and Building 18. The route for the outdoor test was chosen so that, starting from building 18, there is initially a 200 m long line-of-sight connection. A

building is then passed through on the first floor. At a distance of around 400 m, the route then passes through a densely built-up area, which represents a non-line-of-sight connection to the transmitter. Following this, the environment of the non-line-of-sight connection changes to a mix of trees and buildings. The longer straight section in this area is around 410 m to 780 m away from the transmitter and is slightly elevated. There is no line-of-sight connection here either. We would take the route to the northernmost point and back in order to record as many samples as possible. Each samples consisted of a received signal strength indicator (RSSI) and signal noise ratio (SNR) values and are shown in Figure 5.

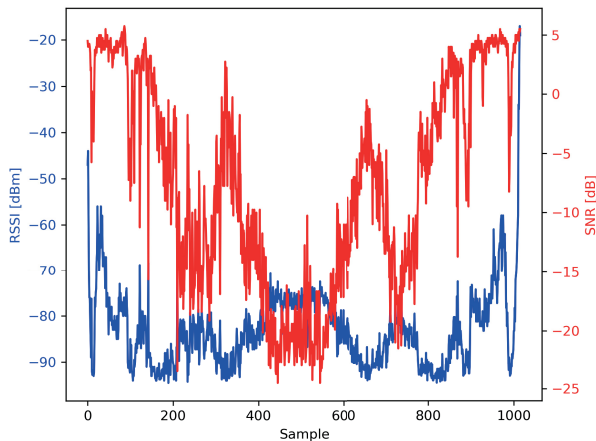


Fig. 5. RSSI and SNR of the Outdoor Range Evaluation with LoRa SF12

Shown in blue is the received signal strength in dBm. As we started the reception strength measurement close to the transmitter, a relatively high received signal strength can be seen at the beginning. We then went into building 18, down the stairwell and back outside. As a result, there is a drop in the received signal strength and an increase. In the middle between 400 and 600 samples, we are about 1 km away from the transmitter at the northernmost point at the top right of Figure 5. We can see that the received signal strength is relatively high, but the signal-to-noise ratio is very low and reception was therefore only possible to a limited extent. This was also shown by the fact that hardly any frames were received in this area, which also represents the limit of the transceivers. According to the resulting data set, the transceiver can receive signals down to -95 dBm in the prevailing radio environment. The signal-to-noise ratio can be as low as -25 dB.

Finally, we compare the results from the model with the range evaluation. In the modeling, we analyzed the technologies using the log-distance path loss model to estimate the range relationship under identical conditions based on modulation and receiver sensitivity. The model indicates that LoRa achieves approximately three times the range of DECT NR+. This estimate was confirmed in the indoor-indoor test, where LoRa reached 56 meters while DECT NR+ achieved 16 meters. However, the model does not account for specific

environmental conditions present in real-world measurements. These factors could be incorporated by replacing the log-distance path loss model with a ray-tracing model, although this is beyond the scope of this publication.

For practical applications, the model-based range estimation provides a useful guideline, but empirical measurements remain essential to determine precise values for specific deployment scenarios.

V. CONCLUSION

This paper evaluated the LPWAN technologies LoRa in the 2.4 GHz band and DECT NR+ in terms of range and data rate across various scenarios. The results indicate that LoRa in the 2.4 GHz band, due to its significant range, is particularly well-suited for applications requiring reliable long-distance communication, such as reading and controlling water meters, as well as transmitting telemetry data for emergency services like firefighters, which exceed the 1% duty cycle limitation of the 868 MHz band. DECT NR+, while limited in range, offers superior data rates, making it ideal for applications requiring higher data rate, such as industrial automation and data-intensive real-time communication.

The combination of modeling and experimental evaluations highlights the key strengths and weaknesses of both technologies. Practical recommendations from this work include using LoRa for scenarios prioritizing robustness and range, while DECT NR+ serves as a valuable alternative for confined areas benefiting from high data rates.

Future research should focus on analyzing additional factors such as interference and mobility. Furthermore, hybrid approaches that combine both technologies could present innovative solutions for IoT applications.

ACKNOWLEDGMENTS

The work is part of funded research project AKTIV (13N16400) of Federal Ministry of Education and Research. Horst Hellbrück is an adjunct professor at the Institute of Telematics of University of Lübeck.

REFERENCES

- [1] P. J. Basford, F. M. J. Bulot, M. Apetroaie-Cristea, S. J. Cox, and S. J. Ossont, "Lorawan for smart city iot deployments: A long term evaluation," *Sensors*, vol. 20, no. 3, 2020. [Online]. Available: <https://www.mdpi.com/1424-8220/20/3/648>
- [2] H. H. R. Sherazi, L. A. Grieco, M. A. Imran, and G. Boggia, "Energy-efficient lorawan for industry 4.0 applications," *IEEE Transactions on Industrial Informatics*, vol. 17, no. 2, pp. 891–902, 2020.
- [3] M. Piechowiak, P. Zwierzykowski, and B. Musznicki, "Lorawan metering infrastructure planning in smart cities," *Applied Sciences*, vol. 13, no. 14, 2023. [Online]. Available: <https://www.mdpi.com/2076-3417/13/14/8431>
- [4] N. Benkahla, H. Tounsi, Y.-Q. Song, and M. Frikha, "Enhanced dynamic duty cycle in lorawan network," in *Ad-hoc, Mobile, and Wireless Networks: 17th International Conference on Ad Hoc Networks and Wireless, ADHOC-NOW 2018, Saint-Malo, France, September 5-7, 2018. Proceedings 17*. Springer, 2018, pp. 147–162.
- [5] A. Samuylov, D. Moltchanov, J. Pirskanen, J. Numminen, Y. Koucheryav, and M. Valkama, "Performance assessment of dect-2020 nr and classic dect coexistence mechanisms," in *2023 IEEE 97th Vehicular Technology Conference (VTC2023-Spring)*. IEEE, 2023, pp. 1–7.

- [6] K. Maliatsos, A. Gotsis, and A. Alexiou, "Evaluation of the imt-2020 candidate radio interface technology dect-2020 nr," in *2022 14th International Conference on COMMunication Systems & NETworkS (COMSNETS)*. IEEE, 2022, pp. 889–893.
- [7] T. Troccoli, H. J. Damsgaard, J. Pirskanen, E. S. Lohan, A. Ometov, J. Morte, J. Nurmi, and V. Kaseva, "Novel direction-of-arrival-based localization in massive dect-2020 5g nr networks," *IEEE Internet of Things Journal*, 2024.
- [8] M. Waßmann, A. Poets, J. Peissig, and J. Pilz, "Performance of dect-2020 nr in an industrial environment for varying rf bands," in *2024 Joint European Conference on Networks and Communications & 6G Summit (EuCNC/6G Summit)*. IEEE, 2024, pp. 610–615.
- [9] M. M. Haque, J. Sae, J. Pirskanen, and M. Valkama, "Dect-2020 nr link distance performance in varying environments: Models and measurements," 2025. [Online]. Available: <https://arxiv.org/abs/2410.24112>
- [10] T. Janssen, N. BniLam, M. Aernouts, R. Berkvens, and M. Weyn, "Lora 2.4 ghz communication link and range," *Sensors*, vol. 20, no. 16, 2020. [Online]. Available: <https://www.mdpi.com/1424-8220/20/16/4366>
- [11] G. H. Derévianckine, A. Guitton, O. Iova, B. Ning, and F. Valois, "Opportunities and challenges of lora 2.4 ghz," *IEEE Communications Magazine*, vol. 61, no. 10, pp. 164–170, 2023.
- [12] Semtech, *Datasheet SX1280 Rev 3.3 2023*.
- [13] SignalVortex. Dect nr mcs index. [Online]. Available: <https://signalvortex.com/dect-nr-mcs-index/>